Chris Lee

Ph.D. Candidate in Operations Research, University of Michigan

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Expected graduation: 05/2029 | **Available:** Summer 2027 (05/2027 – 08/2027)

Research Interests

Online learning and sequential decision-making under structured feedback; bandit algorithms; Bayesian models of sequential decisions; applications in workforce management, energy, and healthcare.

Education

University of Michigan **09/2024 – 05/2028 (expected)**
Ph.D. in Operations Research, Department of Industrial and Operations Engineering Ann Arbor, MI
Advisors: Xiuli Chao and Izak Duenyas. Research area: online learning and stochastic bandits.

University of Washington **2018 – 2022**
B.S. in Mathematics with Honors Seattle, WA
GPA: 3.7. Distinctions: Departmental Honors; Dean's List (2018–2022).
Honors thesis: *Computing Mixing Times of Parallel Tempered Markov Chains.* Advisors: Soumik Pal and Rekha Thomas.

Industry Experience

AmFam Group (American Family Insurance) **08/2022 – 08/2024**
Cloud Data Engineer Madison, WI (remote)

- Lead architect and developer, on a five-person team, of a 12-month initiative to automate key parts of the actuarial ratemaking process over terabyte-scale data (Google Cloud Storage, Cloud Functions, BigQuery, Cloud Composer, Airflow, Google Workspace).
- Built a business-intelligence semantic layer in BigQuery in collaboration with data governance engineers, data scientists, data analysts, and project managers (Cloud Composer, Airflow, Python, SQL).
- Developed an internal tool for evaluating data quality of terabyte-scale tables.
- Received the 2023 "Boomtown Developer" award from the AmFam IT organization for outstanding achievement and passion for learning.
- Rated "exceeds expectations" after three months and "outstanding" after one year in performance reviews.

American Family Insurance **06/2021 – 08/2021**
Data Analytics Intern Madison, WI (remote)

- Produced monthly KPI reports and ad hoc analyses for the performance management department of AmFam Personal Auto lines: pulled data from GCP with SQL, analyzed in Excel and R, and presented findings in PowerPoint.
- Trained a regression model to estimate key partners' mix of business (R).
- Researched and pitched a nonprofit to AmFam's Philanthropy team, resulting in a $500 grant.
- Rated as "exceeded expectations" and a "fast learner" in performance review.

Market Games **09/2020 – 12/2020**
Marketing Analytics and Data Science Intern Berkeley, CA

- Independently built and managed a social media marketing data pipeline: batch ingestion of relational and non-relational sources, including a web scraper for competitive analysis (Python), joined with Google Analytics data.
- Delivered weekly reports with actionable insights for a non-technical team, using clustering for customer segmentation, sentiment analysis of customer interactions, and regression models to optimize marketing variables (R).
- Increased Instagram followers by 30% and likes per post by about 90%.

Research Experience

Washington Experimental Mathematics Lab, University of Washington **09/2021 – 12/2021; 03/2022 – 06/2022**
Undergraduate Researcher, Department of Mathematics Seattle, WA

- Conducted research on graph boundary structure, Markov-chain mixing, graphical designs, and spectral graph theory through an honors thesis, the Washington Experimental Mathematics Lab, and the Washington Directed Reading Program.
- Co-authored a paper published in the *Journal of Combinatorics*; research acknowledged in a paper on graphical designs and Gale duality.
- Faculty mentors: Stefan Steinerberger, Rekha Thomas, and Soumik Pal; graduate mentors: Kevin Liu and Catherine Babecki.

CocoBot, UW School of Public Health **06/2019 – 12/2019**
Research Assistant Seattle, WA

- Prototyped a conversational healthcare AI with Rasa NLU as part of a funded research project (Python).
- Scraped websites for medical diagnostic criteria and treatments and ingested them into Elasticsearch.

Publications and Working Papers

Working Papers

1. **Chris Lee**, Xiuli Chao, and Izak Duenyas. "Sequential Hiring of Contingent Workers Through Learning-Based Optimization." *Submitted to Operations Research*, 2026.
2. **Chris Lee**, Xiuli Chao, and Izak Duenyas. "The Price of Shared Shocks: From Coverage to Transport in Combinatorial Semi-Bandits." *Working paper*, 2026.

Refereed Publication

1. Nick Chiem, William Dudarov, **Chris Lee**, Sean Lee, and Kevin Liu. "A Characterization of Graphs with at Most Four Boundary Vertices." *Journal of Combinatorics*, 15(3), 2024. arXiv: 2209.04438. *Authors listed alphabetically.*

Acknowledged In

- "Graphical Designs and Gale Duality." arXiv: 2204.01873.

Teaching Experience

Project Management **2025, 2026**
Teaching Assistant, University of Michigan Ann Arbor, MI

Linear Programming (Graduate Level) **2026**
Teaching Assistant, University of Michigan Ann Arbor, MI

Advanced Linear Algebra Tools and Applications **01/2022 – 03/2022**
Undergraduate Teaching Assistant, University of Washington Seattle, WA

Selected Projects

- Backend for an airline flight search and ticketing application (Java, SQL).
- Microtargeting system for political advertisements (R, HTML).
- Breaking Google CAPTCHA with machine vision (Python).
- AWS Athena to BigQuery SQL conversion tool (SQL, Python, regex).
- Multi-platform data profiling (Python, SQL, GCP, Excel).
- Home automation with smart appliances (Python).

Skills

- **Methods:** online learning, multi-armed and combinatorial semi-bandits, sequential decision-making, Bayesian modeling, optimization, linear programming, Markov chains and Monte Carlo methods, spectral graph theory, regression, clustering, sentiment analysis.
- **Programming:** Python (pandas, scikit-learn, PySpark, TensorFlow), R (dplyr, tidyverse, ggplot2), SQL (BigQuery, Athena), Java, regular expressions.
- **Data and cloud:** Google Cloud Platform, Amazon Web Services, Apache Airflow, Apache Spark, Docker, Git, GitLab; ETL, data warehousing, data lakes, database design and normalization, star-schema data modeling, data governance; NoSQL (DynamoDB, Elasticsearch, Neo4j).
- **Project management:** business requirements documents, source-to-target mapping, scoping, release cadence, Agile development (Jira, Confluence); cross-functional collaboration with engineers, data scientists, analysts, and project managers.
- **Tools:** Excel (pivot tables, XLOOKUP), PowerPoint.

Awards

- "Boomtown Developer," American Family Insurance IT, 2023.
- 1st Place, UW CSE 416 Kaggle Competition, 2020 (Python).
- 1st Place, Best Data Analysis, UW Datathon, 2019 (R, Excel, PowerPoint).
- 1st Place, UW CSE Bot Tournament, 2018 (Java).
- Data Science Certificate, DubsTech.